

Prompt input/output reference — inputs, filled prompt, and output per call

Every LLM call in the current design, in pipeline order. For each: an **Inputs** list (field → what it is), the **filled prompt** the model receives (template variables substituted with data), and the **output** (structured-output schema + example).

Two conventions, locked from the EDA/experiments:

- Every generation and selection prompt reads the **raw text** (the ticket's consolidated content, ~24k tokens). It is **not** a summary. In the prompts this content sits in a slot the code still labels `ideaCard` / `IDEA CARD SUMMARY` for legacy reasons — the *content* is raw text; the label is stale.
- The **summary** is used **only** as the embedding query to *find* the 6 similar past tickets (retrieval). It never goes into a generation prompt. ("Summary to find, raw to decide.")

To avoid repeating the 24k blob, the raw text is shown once and referenced as `{raw text}` below:

```
{raw text} = "Enabling Real-Time Quote Automation for Enterprise Accounts. Sales Operations needs a real-time CPQ integration so enterprise quotes go out same-day instead of taking 3-5 days. Pricing comes live from Oracle ERP via Salesforce CPQ. Discounts above 20% route to VP approval. Accepted quotes hand off to order fulfilment; the Deal Desk SLA target is 4 hours. [...full consolidated ticket content...]"
```

1. Condense (LLM ×1, per ticket)

Inputs:

- `ticketId` → the ticket id, for traceability.
- `consolidatedText` → the full raw text (description + extracted attachments, ~24k tokens).

Filled prompt (user message):

```
Ticket ID: IDMT-19761
```

```
Source material:  
{raw text}
```

Output — `SummaryFields`:

```
{
  "generatedSummary": "Sales Operations needs a real-time CPQ integration to cut the enterprise quote cycle from five days to same-day, with automated discount approval and a handoff to order fulfilment.",
  "businessProblem": "Manual quoting delays enterprise deal closures by 3-5 days.",
  "businessCapability": "Automated real-time quote generation for enterprise accounts.",
  "keyTerms": ["CPQ", "quoting", "enterprise", "Salesforce", "Deal Desk"],
  "stakeholders": ["Sales Ops", "IT", "Finance"],
  "systemsAndProducts": ["Salesforce CPQ", "Oracle ERP", "Deal Desk Portal"]
}
```

The summary fields are used for **retrieval** (embedding query) and routing — not fed into the generation prompts below.

2. Value Stream Selection (LLM ×1, per ticket)

Inputs:

- `content` → the new ticket's **raw text** (what the model reads to decide). *(The summary is used separately as the embedding query to fetch the 6 past tickets — it is not in this prompt.)*
- `requestedCount` → how many Value Streams to return (exact; default 10).
- `historicEvidence` → the **6 similar past tickets**, each shown as its summary + the Value Streams it was tagged with (precedent).
- `candidateBlocks` → **all 50 governed Value Streams**, one compact block each.

Filled prompt (user message):

IDEA CARD SUMMARY: ← legacy label; the content below is the raw text

{raw text}

REQUESTED VALUE STREAM COUNT (exact):

10

SIMILAR PAST TICKETS (evidence – the value streams these were tagged with):

– IDMT-17432: CPQ enhancements for the large Deal Desk; automated pricing configuration for enterprise tiers.

→ tagged value streams: Configure, Price and Quote (VSR-0042), Order Management (VSR-0017)

– IDMT-16890: Order fulfilment integration with CPQ; quote-to-order handoff latency reduction.

→ tagged value streams: Order Management (VSR-0017)

CANDIDATE VALUE STREAMS:

Candidate: Configure, Price and Quote

entity_id: VSR-0042

description: Initiate, price and issue enterprise quotes.

category: Sales and Enrollment

trigger: A qualified enterprise opportunity needs a quote.

value: Faster, accurate quotes that shorten the sales cycle.

assumptions: Pricing is sourced from the master pricing system.

Candidate: Order Management

entity_id: VSR-0017

description: Capture and fulfil enterprise orders.

category: Fulfilment

trigger: An accepted quote becomes an order.

value: Reliable, on-time order fulfilment.

assumptions: Orders originate from approved quotes.

[... 48 more candidate blocks, all 50 governed Value Streams ...]

Output — recommendations[]:

```
{
  "recommendations": [
    { "valueStreamId": "VSR-0042", "valueStreamName": "Configure, Price and Quote", "confidence":
91,
      "supportType": "direct", "reason": "Ticket explicitly targets CPQ automation for enterprise
quoting.", "sourceTickets": [] },
    { "valueStreamId": "VSR-0017", "valueStreamName": "Order Management", "confidence": 74,
      "supportType": "implied", "reason": "Quote-to-order handoff implied by the fulfilment
requirement.", "sourceTickets": ["IDMT-16890"] }
  ]
}
```

Human approval gate — the SME confirms the Value Stream set before any generation runs.

3. Stage Selection (LLM ×1, all Value Streams batched)

Inputs:

- `content` → the ticket's raw text.
- `valueStreams` → each approved Value Stream with its **own candidate stages**; each stage block is `[sequence] Stage Name (stageId) + description + entrance/exit criteria`.

How the prompt tells the model to read stages: match the work's concrete **action** to a stage's **scope** — its description, entrance and exit criteria, value items, stakeholders — *not* on stage-name similarity; return only stage ids printed under that Value Stream; no count cap.

Filled prompt (user message):

```
## Ticket context
- content: {raw text}

## Approved value streams (each with its own candidate stages)
### Value Stream VSR-0042 – Configure, Price and Quote
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote.
entrance: opportunity qualified | exit: quote issued
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order.
entrance: quote accepted | exit: order created
[3] Quote Revision (VSS-0042-03)
description: Revise an issued quote on customer request.
entrance: revision requested | exit: revised quote issued

### Value Stream VSR-0017 – Order Management
[1] Order Capture (VSS-0017-01)
description: Receive and validate the enterprise order.
entrance: order requested | exit: order validated
[2] Order Fulfilment (VSS-0017-02)
description: Pick, pack and ship the order.
entrance: order confirmed | exit: order shipped
```

The stage **name** is in each block (`[1] Opportunity to Quote (VSS-0042-01)`), so the model matches on scope and returns the stageId.

Output — one entry per Value Stream:

```
{
  "value_streams": [
    { "valueStreamId": "VSR-0042", "selectedStages": [
      { "stageId": "VSS-0042-01", "stageName": "Opportunity to Quote", "reason": "Ticket targets
the quoting initiation workflow." },
      { "stageId": "VSS-0042-02", "stageName": "Quote to Order", "reason": "Quote-to-order
handoff is in scope." } ] },
    { "valueStreamId": "VSR-0017", "selectedStages": [
      { "stageId": "VSS-0017-01", "stageName": "Order Capture", "reason": "Accepted quotes are
captured as orders." } ] }
  ]
}
```

A stage placed under the wrong Value Stream is salvaged to its owner. Empty list = "take the whole governed list for the architect to trim."

4. Description BODY (LLM ×1, per ticket, VS-agnostic)

Inputs:

- `content` → the ticket's raw text.

System: write the shared narrative body; every statement must trace to a phrase in the content; availability/plan lines only when the content explicitly states them.

Filled prompt (user message):

```
Ticket context:
- content: {raw text}
```

Output — { text }:

```
This theme automates enterprise quoting by integrating Salesforce CPQ with live Oracle ERP
pricing,
cutting the quote cycle from five days to same-day. Discounts above 20% route to VP approval, and
accepted quotes hand off to order fulfilment within the Deal Desk's 4-hour SLA.
```

5. Description FRAMING (LLM ×1, all Value Streams batched)

Inputs:

- `content` → the ticket's raw text.
- `valueStreams` → each approved Value Stream: `valueStreamId`, `valueStreamName`, `valueStreamDescription`, `valueProposition`.

Filled prompt (user message):

Ticket context:

– content: {raw text}

Approved value streams:

– valueStreamId: VSR-0042

valueStreamName: Configure, Price and Quote

valueStreamDescription: Initiate, price and issue enterprise quotes.

valueProposition: Faster, accurate quotes that shorten the sales cycle.

– valueStreamId: VSR-0017

valueStreamName: Order Management

valueStreamDescription: Capture and fulfil enterprise orders.

valueProposition: Reliable, on-time order fulfilment.

Output — framings[] :

```
{
  "framings": [
    { "valueStreamId": "VSR-0042", "text": "Within Configure, Price and Quote, this initiative makes enterprise quoting real-time and rule-driven." },
    { "valueStreamId": "VSR-0017", "text": "For Order Management, it ensures accepted quotes flow cleanly into fulfilment." }
  ]
}
```

Final per-VS description = its framing paragraph + the shared body, concatenated deterministically.

6. Business Needs (LLM x1 per Value Stream)

Inputs:

- valueStream → valueStreamId, valueStreamName, valueStreamDescription, valueProposition.
- selectedStages → that VS's selected stages ([seq] Stage Name (stageId) + description + criteria).
- content → the ticket's raw text.

System: write the consolidated Business Needs; every need, dependency, and rule must trace to a phrase in the content; one "Value Stage:" block per stage.

Filled prompt (user message):

```
## Approved value stream
ID: VSR-0042
Name: Configure, Price and Quote
Description: Initiate, price and issue enterprise quotes.
Value proposition: Faster, accurate quotes that shorten the sales cycle.

## Selected stages (write needs for these stages only)
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote. entrance: opportunity qualified | exit: quote issued
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order. entrance: quote accepted | exit: order created

## Ticket context
- content: {raw text}
```

Output — { text } (one consolidated document):

Value Stage: Opportunity to Quote

Business Product Feature: Real-Time Quoting

1. Sales Operations requires real-time quote generation to replace the current 3-5 day manual cycle.

Dependency: Salesforce CPQ and the Oracle pricing API.

Business Rule: Discounts above 20% require VP approval.

Value Stage: Quote to Order

Business Product Feature: Order Handoff

1. Accepted quotes must hand off to order fulfilment within the Deal Desk 4-hour SLA.

7. Capabilities (L3) (LLM x1, ALL Value Streams merged)

Inputs:

- content → the ticket's raw text.
- valueStreams → grouped **Value Stream** → **Stage** → **governed candidate L3**. Each candidate L3 line is capabilityId | name - description [tier] (L2: parent).

System: for each stage, select L3 only from THAT stage's printed candidate list; strict stage isolation (the same id/name repeats across stages and Value Streams — match the exact printed id only). **L2 is derived 1-1 from the selected L3 — no separate L2 call.**

Filled prompt (user message):

```

## Ticket context
- content: {raw text}

## Approved value streams, each with its selected stages and each stage's own candidate L3
### Value Stream VSR-0042 – Configure, Price and Quote
description: Initiate, price and issue enterprise quotes.

### Stage VSS-0042-01
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote. entrance: opportunity qualified | exit: quote issued
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0042-0011 | Real-Time Pricing Engine Integration – live CPQ pricing feed [tier: core] (L2: Pricing and Discounting)
- L3-0042-0012 | Automated Discount Approval Workflow – routes discounts for approval (L2: Approval and Governance)

### Stage VSS-0042-02
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order. entrance: quote accepted | exit: order created
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0042-0021 | Order Handoff Orchestration – passes the quote to fulfilment (L2: Order Management)

### Value Stream VSR-0017 – Order Management
description: Capture and fulfil enterprise orders.

### Stage VSS-0017-01
[1] Order Capture (VSS-0017-01)
description: Receive and validate the enterprise order. entrance: order requested | exit: order validated
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0017-0011 | Order Intake Automation – captures order data (L2: Order Management)

```

Output — keyed by stageId (re-attributed to each Value Stream afterward):

```

{
  "stages": [
    { "stageId": "VSS-0042-01", "capabilities": [
      { "capabilityId": "L3-0042-0011", "capabilityName": "Real-Time Pricing Engine Integration", "reason": "Live CPQ pricing feed from Oracle ERP." },
      { "capabilityId": "L3-0042-0012", "capabilityName": "Automated Discount Approval Workflow", "reason": "Discounts above 20% require VP approval." } ] },
    { "stageId": "VSS-0042-02", "capabilities": [
      { "capabilityId": "L3-0042-0021", "capabilityName": "Order Handoff Orchestration", "reason": "Accepted quotes hand off to fulfilment." } ] },
    { "stageId": "VSS-0017-01", "capabilities": [
      { "capabilityId": "L3-0017-0011", "capabilityName": "Order Intake Automation", "reason": "Captures the order on quote acceptance." } ] }
  ]
}

```

*Strict isolation + salvage keep each L3 under its owning stage. **L2** = the unique parent L2s of the selected L3, derived per Value Stream.*

Final Theme package (deterministic — no LLM)

One package per approved Value Stream:

```
{
  "themeTitle": "Enabling Real-Time Quote Automation -- Configure, Price and Quote",
  "themeDescription": "<framing + body>",
  "selectedStages": [ { "stageId": "VSS-0042-01", "stageName": "Opportunity to Quote", "reason":
"..."} ],
  "businessNeeds": "<consolidated text>",
  "l3Capabilities": [ { "stageId": "VSS-0042-01", "capabilities": [ ... ] } ],
  "l2Capabilities": [ { "stageId": "VSS-0042-01", "capabilities": [ { "capabilityId": "L2-0042-
001", "name": "Pricing and Discounting" } ] } ]
}
```

Evaluation judges (eval-only, reference-free)

Never run in production. They judge a generated artifact against its **source** (the raw text), not ground truth. Claim extraction runs once; its claims feed faithfulness and correctness.

J1. Claim extraction • `judges/claim_extraction`

Inputs: `text` → the generated artifact. **Filled prompt:**

GENERATED TEXT:

This theme automates enterprise quoting by integrating Salesforce CPQ with live Oracle ERP pricing, cutting the quote cycle from five days to same-day. Discounts above 20% route to VP approval, and accepted quotes hand off to order fulfilment.

List every atomic factual claim.

Output — `ClaimList`:

```
{ "claims": [
  "The integration cuts the enterprise quote cycle from five days to same-day",
  "Discounts above 20% route to VP approval",
  "Accepted quotes hand off to order fulfilment" ] }
```

J2. Faithfulness / hallucination • `judges/faithfulness`

Inputs: `source` → the raw text; `claims` → the extracted claims. **Filled prompt:**

```
SOURCE:
{raw text}

CLAIMS:
- The integration cuts the enterprise quote cycle from five days to same-day
- Discounts above 20% route to VP approval
- The feature launches in Q3

Return each claim with its supported flag.
```

Output — `FaithfulnessResult` (faithfulness = supported/total; hallucination = 1 – it):

```
{ "claims": [
  { "claim": "...quote cycle from five days to same-day", "supported": true },
  { "claim": "Discounts above 20% route to VP approval", "supported": true },
  { "claim": "The feature launches in Q3", "supported": false } ] } // -> faithfulness 0.67
```

J3. Correctness • `judges/correctness`

Inputs: `source` + `claims` (same shape as faithfulness). Checks accuracy / no distortion. **Output** — `CorrectnessResult` (correct/total):

```
{ "claims": [
  { "claim": "Discounts above 20% require VP approval", "correct": true },
  { "claim": "All discounts require VP approval", "correct": false } ] } // scope distorted
```

J4. Coverage • `judges/coverage`

Inputs: `source` → the raw text; `description` → the generated artifact. **Filled prompt:**

```
SOURCE:
{raw text}

GENERATED ARTIFACT:
This theme automates enterprise quoting ... cutting the quote cycle to same-day.

Extract the source's key facts and return each with whether the generated artifact covers it.
```

Output — `CoverageResult` (covered/total):

```
{ "facts": [
  { "fact": "Cut quote cycle from 5 days to same-day", "covered": true },
  { "fact": "Hand off accepted quotes to fulfilment", "covered": false } ] } // -> coverage 0.5
```

J5. Stage usage (Business Needs only) · `judges/stage_usage`

Inputs: `stages` → the selected stages with scope; `business_needs` → the generated document. **Filled prompt:**

```
SELECTED STAGES (with their scope):
[1] Opportunity to Quote (VSS-0042-01)  description: Initiate and price a quote. entrance: ... |
exit: quote issued
[2] Quote to Order (VSS-0042-02)        description: Convert an accepted quote into an order. ...

BUSINESS NEEDS DOCUMENT:
Value Stage: Opportunity to Quote ... Value Stage: Quote to Order ...

For each selected stage, return addressed + aligned + a short note.
```

Output — `StageUsageResult` (usage = addressed/selected; alignment = aligned/addressed):

```
{ "stages": [
  { "stageId": "VSS-0042-01", "addressed": true, "aligned": true, "note": "Real-time quoting needs
are in scope." },
  { "stageId": "VSS-0042-02", "addressed": true, "aligned": false, "note": "An order-creation need
is filed here but belongs to a later stage." } ] }
```